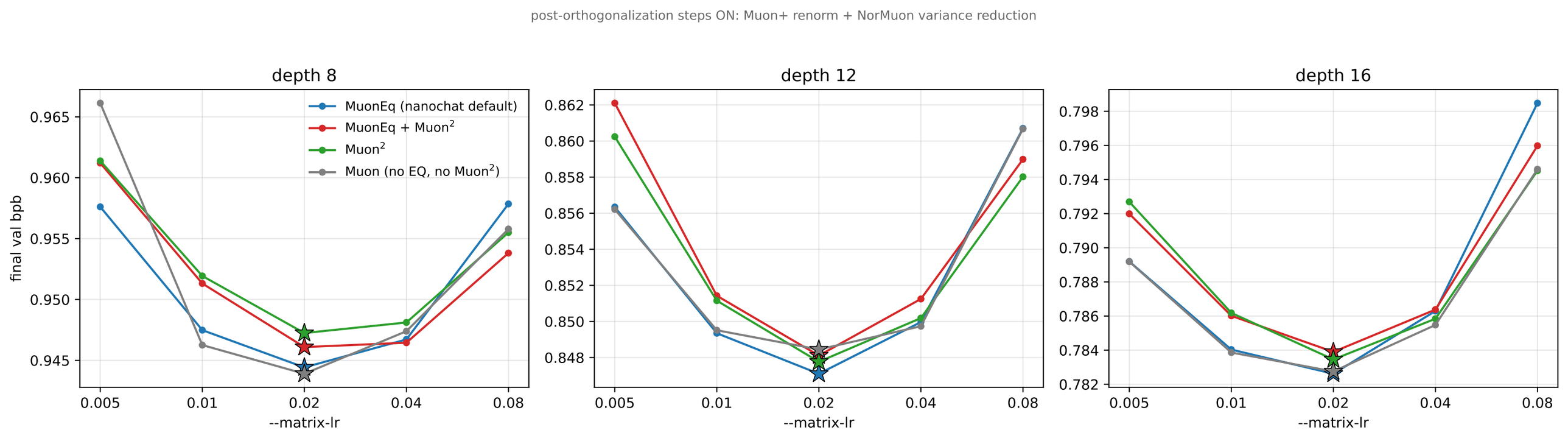
post-orthogonalization steps ON: Muon+ renorm + NorMuon variance reduction
depth 8
depth 12
depth 16
MuonEq (nanochat default)
MuonEq + Muon²
Muon²
Muon (no EQ, no Muon²)
final val bpb
--matrix-lr
0.005
0.01
0.02
0.04
0.08
0.945
0.950
0.955
0.960
0.965
0.848
0.850
0.852
0.854
0.856
0.858
0.860
0.862
0.782
0.784
0.786
0.788
0.790
0.792
0.794
0.796
0.798